Improving video call quality on home and office networks

Article KB0010027 | Owner: Network Support | Subcategory: Slow network | Applies to: Windows 11 laptops (Dell Latitude, Lenovo ThinkPad, HP EliteBook), macOS 14/15 laptops (MacBook Pro/Air), Mobile devices using Teams/Outlook, Desktop workstations connected via Ethernet or Wi-Fi

Summary

This article helps users diagnose and improve video-call quality on both home and office networks. It outlines common symptoms, typical causes, quick self-checks you can perform, and step-by-step guidance for optimizing Wi-Fi, wired connections, and application settings. Service-desk agents can follow a separate procedure to investigate congestion, AP load, and ISP issues.

Symptoms

- Video freezes or pixelates during Teams/Zoom calls
- Audio sounds choppy or echoey
- Call drops or reconnects frequently
- Measured latency >150?ms or jitter >30?ms on speed test
- Download/upload speeds far below contracted ISP plan

Common causes

- Wi-Fi congestion on the 2.4?GHz band or overloaded access point
- Device connected to a weak signal or far from AP
- QoS/prioritization disabled on router or switch
- Background applications consuming bandwidth (cloud sync, large downloads)
- VPN split-tunneling routing traffic through congested paths
- ISP throttling or local outage
- Outdated network adapter drivers or firmware

Overview

Video-call quality depends on sufficient bandwidth, low latency, and minimal jitter. Most issues stem from the local wireless network (home router or office AP) rather than the ISP. By checking signal strength, band selection, and competing traffic, users can often restore smooth calls without needing a ticket.

Quick Self-Help Checks

Follow these steps to verify and improve your connection before contacting the Service Desk. They take less than five minutes and resolve the majority of home-office video-call problems.

1. Run a speed test: open https://speedtest.acme.example (or any trusted site) and note download, upload, ping, and jitter. Compare to your ISP plan.
2. Check Wi-Fi signal: click the network icon -> look for signal bars. If <=2 bars, move closer to the router or AP.
3. Switch to 5?GHz band: in Wi-Fi settings, ensure you are connected to the SSID ending with "_5G" or similar; avoid the 2.4?GHz band for video.
4. Pause bandwidth-heavy apps: stop OneDrive sync, large file downloads, or streaming video while on the call.
5. Enable QoS (if your router supports it): log into the router admin page, locate WMM/MM or QoS settings, and prioritize video-conferencing traffic (TCP/UDP ports 443, 3478-3481, 50000-50019).
6. Restart the router/AP: power-cycle for 30?seconds to clear temporary glitches.

Improving Your Home Network

If the quick checks did not resolve the issue, consider these deeper actions that target common home-network bottlenecks.

1. Upgrade to a dual-band or Wi-Fi?6 router if you are still using an older single-band device.
2. Place the router in a central location, away from thick walls, microwaves, or cordless phones.
3. Use a wired Ethernet connection for the laptop whenever possible; a USB-C to Ethernet adapter works on most modern laptops.

4. Check for firmware updates on the router manufacturer's website and apply them.
5. Limit the number of simultaneous devices streaming 4K video or gaming during work hours.

Improving the Office Network

In an Acme office, video-call degradation is usually caused by AP overload or channel interference. The steps below can be performed by the user or reported to the Network Support team.

1. Confirm you are connected to the SSID "ACME-Corp" (802.1X) and not the guest network.
2. Ask a colleague nearby if they experience the same issue; if yes, the AP may be congested.
3. If possible, move to a different desk or conference room served by another AP.
4. Check the Teams call statistics: in-call -> Settings -> Device -> Call quality -> look for "Network" metrics; high packet loss indicates AP issues.
5. Report persistent problems to the Service Desk (phone?4357) with the AP name (shown in the Wi-Fi details) and the time of day.

Verification

After applying changes, verify that video-call quality has improved before closing the ticket or considering the issue resolved.

1. Make a test Teams call with a colleague and use the in-call diagnostics (Settings -> Device -> Call quality).
2. Ensure jitter <30?ms, packet loss <1?%, and average latency <150?ms.
3. Run a second speed test to confirm bandwidth meets the minimum 3?Mbps upload/download for HD video.
4. If quality remains poor, proceed to escalation.

Service Desk Procedure

Agents follow this workflow when a user reports video-call slowness that persists after self-help.

1. Gather baseline information: user location (home/office), device type, OS version, time of day, and whether the issue affects multiple users.
2. Check recent network alerts in the monitoring dashboard for AP utilization, channel interference, or controller alarms.
3. If the user is on Wi-Fi, request the connected AP name and signal strength (RSSI) from the user or via Intune device details.
4. For home users, advise to run a speed test and share results; note any ISP-reported outages on the status page.
5. For office users, verify AP load: if >60?% client count or >70?% airtime utilization, initiate AP load-balancing or channel change via the Wireless LAN controller.
6. If the problem is isolated to a single device, check driver/firmware version and recommend updating the network adapter.
7. Document all findings, steps taken, and user feedback in the ticket before closing.

When to escalate

Escalate to the Network Support team when: (a) multiple users on the same floor or building report simultaneous video-call degradation, (b) AP utilization exceeds 80?% or channel interference alerts are present, (c) speed tests show ISP-provided bandwidth consistently below the contracted tier, or (d) the issue persists after AP load-balancing, firmware updates, and wired-connection testing. Include in the escalation: ticket number, affected SSID/AP names, timestamps, speed-test results, number of impacted users, and any relevant controller logs.

Keywords: video call quality, Teams lag, Zoom freezing, slow Wi-Fi, 5?GHz vs 2.4?GHz, AP congestion, QoS, network jitter, speed test, router restart, wired connection, office Wi-Fi, home network troubleshooting